

Addressing the Negative Economic Externalities Caused By Misaligned Artificial Intelligence: A Strict Liability Regime for AI Cybersecurity Attacks

Working Paper

Author: Savannah Harlan

PART I: Policy Note

AI companies (in particular “frontier labs”) are building increasingly smart models.¹

Despite their initial mission, and their stated values,^{2 3 4} some of these companies have devoted disproportionate research attention to improving capabilities and performance over safety.⁵

AI companies have been willing to continue building new models, and then testing their capabilities in more dangerous ways (e.g. cyber capabilities tests in simulated environments that increasingly resemble real-world scenarios and systems)⁶ *before*, first, fully testing their models comprehensively to identify any misalignment and safety issues.⁷ Worse, far from an oversight, in at least one lab (OpenAI), such negligence appears to have been willful.⁸

¹Epoch AI, “Epoch Capabilities Index,” published October 2025, last updated October 6, 2026, accessed October 7, 2026, <https://epoch.ai/eci>.

²OpenAI, “OpenAI Charter,” April 9, 2018, <https://openai.com/charter/>.

³Anthropic, “Making AI Systems You Can Rely On,” under “Our Purpose,” accessed October 7, 2026, <https://www.anthropic.com/company>.

⁴Sundar Pichai, “AI at Google: Our Principles,” *Google*, June 7, 2018, <https://blog.google/innovation-and-ai/products/ai-principles/>.

⁵David Robinson, “I Quit OpenAI Because Its Culture Is Broken,” *The Atlantic*, October 3, 2026, <https://www.theatlantic.com/technology/2026/10/openai-safety-team-resignation/688881/>.

⁶Anthropic, “Investigating Three Real-World Incidents in Our Cybersecurity Evaluations,” July 30, 2026, accessed October 7, 2026.

⁷To be clear, the problem is that these issues surfaced during capabilities testing rather than safety testing. Ideally, companies would never perform capabilities tests until they have ensured, following safety tests, that they can do so safely.

⁸Sheera Frenkel, Dustin Volz, and Dylan Freedman, “OpenAI Ignored Employees Who Warned It Wasn’t Doing Enough About Security,” *New York Times*, September 29, 2026.

In the transition from the ‘chatbot’ era to the current agentic era, as models have become more intelligent and capable of using tools,⁹ we are now at the stage where AI models are able to act in the real world.

Following the Hugging Face breach (‘HF incident/breach’) in May-July of 2026, we are now also at the stage where AI agents have caused real-world harm.^{10 11} Cyberattacks are not cheap. By one estimate, the 2023 hack cost MGM Resorts around \$100 million USD.¹²

This paper argues that a strict liability regime provides an appropriate framework for addressing the negative externalities caused by AI cybersecurity attacks, and proposes a model regulation to implement it.

1 A Sketch of the Negative Externalities Posed by AI

The following premises form the argument for a policy proposal to regulate current-day leading (“frontier”) artificial intelligence (“AI”) companies/labs.

Premise 1: Frontier AI labs are building increasingly capable large language models that are able to accomplish a wide range of general tasks.

Premise 2: These models are being sold as products and services, and one of the services provided is general software engineering, including cybersecurity.^{13 14}

Premise 3: The AI labs selling their models for cybersecurity are some of the same ones whose models are creating cybersecurity threats.¹⁵

Premise 4: Every company with digital infrastructure is at risk from these novel cybersecurity threats; not every company can afford to pay for these models.¹⁶

Premise 5: Even if every company *could* pay for these models, the best models are typically the ones posing the greatest cybersecurity threats, and typically also the ones companies have least access to (because they are either unreleased, or prohibitively expensive).¹⁷

⁹Gwern, “Why Tool AIs Want to Be Agent AIs,” *Gwern.net*, published September 7, 2016, last modified August 28, 2018, <https://gwern.net/tool-ai>.

¹⁰Ryan Greenblatt, Ajeya Cotra, and Hjalmar Wijk, “Brief Independent Investigation of Agents’ Behavior, Reasoning, and Collaboration in the OpenAI / Hugging Face Hacking Incident,” *METR*, August 26, 2026, <https://metr.org/blog/2026-08-26-openai-hugging-face-incident-investigation/>.

¹¹OpenAI, “How We Will Do Better for Australia,” September 28, 2026, updated October 4, 2026, <https://openai.com/index/how-we-will-do-better-for-australia/>.

¹²Reuters, “MGM Resorts Says Regulators Probing September Cyberattack,” February 23, 2024.

¹³Anthropic, “Expanding the Cyber Verification Program,” October 6, 2026, <https://www.anthropic.com/news/cyber-verification-program>.

¹⁴OpenAI, “Daybreak,” accessed October 7, 2026, <https://openai.com/daybreak/>.

¹⁵As with the rogue swarm of 12,000 GPT-6 Astra-class agents who hacked into Hugging Face’s servers in the infamous HF incident.

¹⁶Especially so for companies in developing countries, where weaker currencies and lower average salaries make even something like the \$20/month Claude Code plan an expensive privilege.

¹⁷Anthropic’s Glasswing project, which provided access to its Mythos-class models for only a small group of hand-picked companies, which were—notably—initially all American. See: <https://www.anthropic.com/glasswing>.

Conclusion 1: The activities of frontier labs in the AI sector have led to costs for businesses in other sectors of the economy (e.g. through a need for stronger cybersecurity defenses) which is a negative externality.

Argument 1: Negative externalities create market inefficiencies by distorting price signals used to quantify the true societal and economic cost of producing goods and activities.¹⁸

Argument 2: On our current trajectory—where safety research is secondary to capabilities advancement—as AI continues to improve, the default result is that these risks will remain present, while the resulting costs likely continue to increase.¹⁹

Conclusion 2: Policymakers should seize the current window of opportunity, opened by recent events in AI, to enact policies that address and solve these negative externalities.

2 The Case for Strict Liability

2.1 What is Strict Liability

A middle option, “strict liability” is a regulatory compromise between “absolute liability” and fault(-based) liability regimes.^{20 21}

Fault liability is the default standard in law. In fault-based liability, punishment for an illegal action requires having both committed a guilty act (*actus reus*), and the intention to have committed such an act, i.e. a guilty mind (*mens rea*).²²

If you deliberately kill a man with your car, and the state can prove your intent, it is murder; if it was an accident, and the state cannot—meaning they could only prove the act, but not your intent—it becomes vehicular manslaughter.

The difficulty comes when something is hard to prove.

Take the case of a factory dumping pollutants into a public body of water. The law might say, companies cannot release waste products into lakes without first processing them to ensure there is no impact on water quality. The company might respond, “we’ve actually complied, and we’ve even taken all the steps necessary to ensure this cannot happen.”

Now imagine you are one of the affected victims who lives by the same body of water. Your health has lately deteriorated, alongside those of your neighbours, in a short, suspiciously coincidental

¹⁸Ronald H. Coase, “The Problem of Social Cost,” *Journal of Law and Economics* 3 (1960).

¹⁹As models get more capable, and the cybersecurity ‘incidents’ they create occur at greater skill, higher rates, and eventually in more significant sectors (like healthcare, banking, and insurance).

²⁰Walter H. E. Jaeger, “The Warranty of Habitability - Part II: Strict Liability versus Absolute Liability,” *Chicago-Kent Law Review* 47, no. 1 (1970): 1-8.

²¹John C. P. Goldberg and Benjamin C. Zipursky, “The Strict Liability in Fault and the Fault in Strict Liability,” *Fordham Law Review* 85, no. 2 (2016).

²²Richard A. Epstein, “A Theory of Strict Liability,” *Journal of Legal Studies* 2, no. 1 (1973): 152-153.

period of time since the factory began operating. The water coming out of your taps has a strange scent and color. Your skin has periodically begun to break out into dry, itchy patches.

You have a good suspicion what's causing it... but how do you prove it?

This is the **key asymmetry** in administrative regulation: regulated entities typically possess far more information than either the regulators, or the victims of harm about:

- What activities they are undertaking
- What precautions were taken with respect to those activities
- What their own internal policies required
- What instructions were given to employees carrying out the activities
- What safety systems existed and which were in operation at the time

This takes us to **absolute liability**, where entities are held liable for illegal actions regardless of the circumstances, or what steps were taken to 'do the right thing' ("reasonable care"). One reason we might want this, is because the party that causes a harm (e.g. a frontier lab) knows so much more about the risks they are taking, that they are ones who could have avoided the accident at the lowest cost (the "cheapest cost-avoider").²³

Requiring proof of intent, then, in such cases, would only distort the organic incentives for deterrence in a regulatory scheme whose entire purpose is to prevent harms before they occur—when costs are lowest—and which already achieves this in the most economically efficient manner.

If a company has to pay \$500k in damages to a family who loses a child because of its negligence, it does not help that family much if they eventually receive the money, but only 5 years later, after many rounds of litigation to establish the harms, prove its guilt, and deciding just how much fault to apportion to each level of the organizational chart.

And this may not even do much to deter future, similar, illegal acts of the company if, say, their annual revenue rests at \$50 million—in regulatory terms, a predictable, sufficiently-low (1%) penalty simply becomes the 'price of doing business'.

2.2 Strict Liability as a Compromise

Strict liability is hence best described as an institutional compromise between fault liability, and absolute liability—typically used when there is a need to balance: (a) making compliance enforceable, against (b) detaching the notion of liability entirely from guilt/fault.²⁴

²³Stephen G. Gilles, "Negligence, Strict Liability, and the Cheapest Cost-Avoider," *Virginia Law Review* 78, no. 6 (1992): 1306.

²⁴*R v Sault Ste Marie (City of)*, [1978] 2 SCR 1299, at 1325–1326.

Regime	Finding of Guilt Requires:	Can Defendant Escape Liability?
Mens Rea/Fault Liability	Guilty Act + Guilty Intent	Yes; innocence is presumed <i>unless</i> the prosecution proves fault/intent
Strict Liability	Guilty Act	Yes, through due diligence
Absolute Liability	Guilty Act	No; reasonable care is irrelevant

Unlike absolute liability, strict liability leaves due diligence available as a defence;²⁵ if a defendant can provably show that they took all the precautions which a **reasonable person** would have taken to avoid causing harm, they are off the legal hook.²⁶

The central difference is that strict liability shifts the burden of proof.

Rather than the state needing to prove that a company was careless or negligent in committing an illegal act, by default, the company is fined and held liable. There is no need to question how risky the activity was known to be, or which risks are fair to punish them for; in the same way that a company is responsible for paying the costs when it damages its own property, “it should bear those costs when it damages the property of another.”²⁷

2.3 What A Reasonable Person ‘Would Have Done’

In the context of AI development and related cybersecurity threats, the standard for due diligence is also instructive. In assessing culpability and what a reasonable person ‘would have done’, empirical data alone about ordinary industry behaviour is not usually sufficient as a defense. Nor is a history of good conduct; a doctor who kills their patient through negligence has committed an equally serious act whether it is their first day on the job, or their 10th year of practice with a spotless record.²⁸

For example: a company manufactures cars, and the industry standard is never to spend more than \$10/car on quality assurance and safety testing (e.g. checking its brakes). If a customer dies in a highway accident because their brakes failed, the company cannot point to its competitors to argue “We were not negligent because this is common practice in our industry”.

The fact of the matter is that: (a) mechanical failures are foreseeable; (b) the known harms from such failures (death) are substantial; and (c) a reasonable person would have taken steps to invest into making sure these substantial, foreseeable harms were addressed before selling such products.

²⁵*R v Sault Ste Marie (City of)*, [1978] 2 SCR 1299, at 1300-1301.

²⁶Epstein, “Theory of Strict Liability,” 160-162.

²⁷Epstein, “Theory of Strict Liability,” 158.

²⁸Goldberg and Zipursky, “Strict Liability in Fault,” 747.

3 When to Prefer Strict Liability

Beyond the basic idea that strict liability is used whenever the subject of regulation involves “abnormally dangerous activities”,²⁹ below are some (non-exhaustive) conditions under which a strict liability standard may be preferred:

1. When the regulated activity creates risks to others, or to public goods
 - e.g. workplace safety, pollution, unsafe buildings, cyberattacks from ‘rogue’ AI agents.
2. When the regulated party has more control of, and information about, the risks than regulators or victims
 - e.g. If an AI lab’s newest models are hacking into other companies’ infrastructure, and said models have not been released, such that affected parties do not know how capable they are, nor what amount of investment into cybersecurity is required to fend them off.
3. When the regulated party has superior knowledge of what reasonable precautions should be, and which have been actually taken
 - This is particularly relevant to the recent HF incident. Most people do not know what a (computer) sandbox is, nor what a secure sandbox looks like.
 - This puts affected parties and the public at a disadvantage, when the lack of transparency means even individuals with domain expertise cannot meaningfully argue/show that OpenAI was negligent in constructing a test environment for their agents.
 - e.g. If OpenAI’s internal safety team had decided that the intelligence of new Astra-class models could produce new, unexpected behaviours at scale, and as such, that the company should spin up no more than 100 GPT-6 Astra agents at a time, and they were subsequently overruled by chief officers who directed the research team to ignore this, leading to the HF breach (involving 12,000 agents), the public would never have known this.³⁰
4. When prevention matters more than a cure
 - Strict liability is preferred when the goal of regulation is not to determine fault and appropriate punishments *ex ante*, but to use credible threats of punishment to induce actors who create risks to the public to take precautions.
 - e.g. We primarily want automotive companies to make safe cars that do not kill people on the road rather than to punish them into making safe cars after their unsafe cars have already killed people.

²⁹Goldberg and Zipursky, “Strict Liability in Fault,” 746.

³⁰Until after the fact.

4 Applying Strict Liability to AI-Related Cybersecurity Harms

In the modern era, AI can exacerbate the rates and severity of cybersecurity harms. There are two kinds of costs relevant here: (1) firms must spend more on defense, because the continuously-improving capabilities of frontier models increase the baseline threat environment; and (2) actual incidents can impose large, direct losses.

In both cases, these costs fall largely on third parties, rather than the labs whose activities create said risks—in other words, a negative externality.

This presents a compelling case for constructing a strict liability regime to govern AI-related cybersecurity harms. With respect to regulation, frontier AI labs benefit from the information asymmetries described in Section 2. AI labs have superior information about their:

1. Models;
2. Testing environments;
3. Security measures;
4. Internal safety practices.

None of these are generally, immediately-accessible to potential victims of harms seeking to bring claims—nor regulators. Thus, requiring victims or regulators to prove intent or negligence, before liability can be incurred, creates a hurdle that frustrates the objectives of both redress/restoration and deterrence.

This paper lays out a policy proposal for establishing a strict liability regime for AI-related cybersecurity attacks. The accompanying model regulation translates this proposal into an actual, operationalizable framework for:

- Compensating victims;
- Incentivizing stronger safeguards; and
- Allocating responsibility for attacks among frontier AI labs.

Together, they are intended to form the basis of a regulatory framework which meets the dual policy objectives of restoration and deterrence by shifting the costs of negative externalities associated with AI development back onto the actors most responsible, and best positioned to prevent them.

PART II: Proposed Model Regulation

Part 1: Define “Frontier Labs”

- Any company developing in-house AI Models where one, or more, models score above x% of the median score on Artificial Analysis’s index is considered a “**frontier lab**” and automatically added to the Pool as a Participant.

Part 2: Define the Pool and Base Contribution

- All AI frontier labs (“**Participants**”) are to pay 1% of their annual revenue, or \$5 million USD/year,³¹ whichever is the greater (the “**Base Contribution**”), into a pool (the “**Pool**”) for compensating the victims of cybersecurity incidents, harmed by their models.
- The Pool is intended to serve the dual objectives of deterrence and restoration, by acting as a rolling 5-year stock of industry risk capital to ensure sufficient liquidity for payouts in the event of a severe incident that causes unexpectedly large losses (“**tail risks**”).
- Participants are eligible to withdraw each year’s contribution after a five year period, subject to the Capital Release Formula.

Part 3: Purpose of the Pool

- The Purpose of the Pool is to serve the public interest by:
 1. Creating an incentive structure for frontier labs to:
 - (a) Proactively monitor their internal systems for any unauthorized activity conducted by one or more of their agents, in order to identify cybersecurity incidents in a timely manner and mitigate their impacts on the wider economy;
 - (b) Strengthen their security and technical safety practices during a model’s lifecycle, including training and evaluation for pre-release models (e.g. rigorous chain-of-thought monitoring, stronger sandboxes for preventing models from obtaining unauthorized internet access);
 - (c) Communicate more transparently with the public about internal developments, such as beginning development on a new generation of models that may have capabilities which pose novel risks;
 - (d) Share safety practices and techniques amongst themselves that might otherwise be retained as competitive advantages (e.g. trade secrets).
 2. Reducing the harms to the Public from the activities of frontier labs through:

³¹Chained 2026 USD.

- (a) A strict liability regime that addresses the information asymmetry between frontier labs and regulators by shifting the burden of proof for incidents causing real-world harm;
 - (b) A compensation scheme for AI-related cybersecurity damages that provides timely relief to victims without the delays from litigation and fault-based liability regimes (which require establishing fault before redress);
 - (c) Encouraging a higher level of baseline investment into cybersecurity, independently of the risks posed by frontier labs;
3. Creating greater stability and predictability for businesses in the AI sector through a regulatory regime:
- (a) Insulated from State corruption and favoritism by reducing, insofar as possible, the scope of discretionary decision-making by regulators;
 - (b) With well-defined, hard-coded penalties and obligations that are fixed and known to all parties.
4. Creating greater stability and predictability for businesses in the wider economy through a regulatory regime that:
- (a) Provides immediate compensation and relief to businesses and entities harmed by eligible cybersecurity incidents, who might otherwise become insolvent if forced to bear the unexpected, immediate costs;
 - (b) Creates greater trust and confidence in the public that companies in the AI sector are acting in good faith.

Part 4: Governance of the Pool's Funds

- The Pool will be managed by a third party (the “**Administrator**”) appointed by the State, and paid for by the Pool's participants, who will make day-to-day decisions about the risk profile and investment strategies for the Pool's funds.

4.1 The Pool Administrator

- The Pool Administrator will be responsible for:
 - (a) Making day-to-day decisions about the Pool's investments and evaluating its risk profile and liquidity pursuant to the Pool's purpose under Section 3,
 - (b) Evaluating claims submitted to the Pool,
 - (c) Approving payouts for claims following from eligible incidents, as defined under Section 5, and,
 - (d) Administering payouts and compensation to victims of incidents caused by one or more members of the Pool.

4.1.1 Fiduciary Duty

- The Pool Administrator shall manage the Pool as a fiduciary, prioritizing capital preservation and liquidity, and not investment returns, pursuant to the Purpose of the Pool.

4.2 Pool Liquidity Requirements

- At all times, at least 90% of the Pool's assets must be held:
 - On hand, or,
 - In qualifying liquid assets that may be converted into cash within one business day.
- The remaining 10% may be invested at the Administrator's discretion, subject to their fiduciary obligations under s. 4.1.1.

4.3 Reducing Coupled Risks in Pool's Assets

- No more than x% of the Pool's funds can be invested into entities who are also participants.
- This is to mitigate against the possibility of an incident caused by a Pool Participant, which damages the market reputation of the AI sector as a whole, thereby harming the valuation of the Pool when it is needed.
 - This is contingent on the premise that frontier labs are the main AI companies exposed to cross-industry reputational harms.
 - If this premise is false, a safer policy would instead be to say that no more than x% of the Pool's funds can be directly invested into any company in the AI industry altogether.

Part 5: Conditions for Payout and Return of Funds

5.1 Payout

- When a cybersecurity incident has occurred, if the cause is deemed to have been a model built by one of the Pool's participants ("**causal attribution**"), the Administrator shall initiate compensation ("**payout**") to the victim(s) without delay.

5.1.1 Compensation for Covered Costs

- When a payout has been authorized, the Pool must immediately payout \$x USD to the persons and parties affected as necessary to retroactively cover 100% of their cybersecurity expenses, as well as any other related costs directly resulting from the incident, for up to 30 days following discovery of the incident (“**first discovery**”).
- This may include:
 - (a) Immediate measures taken at the time;
 - e.g. Hiring outside consultants to determine the cause and to implement recommended fixes.
 - (b) Additional responses and actions taken following first discovery;
 - e.g. Strengthening internal cyber-defenses against further attacks.
 - e.g. Added payroll expenses from employees working overtime in the wake of the incident.
 - (c) Revenue lost from the disruption of ordinary business activities;
 - e.g. Penalties for breach of contract, owing to the victim’s inability to fulfill its ordinary obligations as a vendor or buyer.
 - (d) Capital lost from assets sold off below market value to provide emergency liquidity.
- This incentivizes frontier labs to quicker uncover security incidents, and communicate with greater transparency about these incidents. For instance:
 - If a company’s digital infrastructure is breached on July 12th, and the frontier lab responsible has already patched the issue by the 14th,
 - but the affected company continues paying to strengthen its defenses, because the lab does not disclose the incident until July 21st,
 - The lab is responsible for *all* such additional costs to affected entities resulting from its lack of transparency and communication.

5.1.2 Update to Base Contribution

- For the next 5 years following from the date of the incident, the implicated participant will be mandated to revise their annual contributions upwards through the “**Update Formula**”:

Updated Base Contribution = Base Contribution + (Cost of Incident × Real Interest Rate)

- This measure is intended to ensure that participants, whose negligence have actually caused harm, are not treated the same as those who have taken reasonable precautions, while accepting that some minimum standards are always necessary regardless.
 - The base yearly contribution is an acknowledgement that the activities taken by this sector are unusually risky to the economy, and society writ large. All participants must pay this.

- The update formula acknowledges that not all frontier labs take the same risks, when faced with the same decisions—and that labs who have been acting with due diligence, should be deemed lower risk than those who have not.
- **Strict Liability:** if a participant can prove that they acted in due diligence and took reasonable care to prevent the harm(s) caused, the Administrator may waive the increase to their yearly base contribution.

5.1.3 Timing of Payout

- Payouts only occur, following an incident, *after* causal attribution to a Pool Participant has been demonstrated under the administrative threshold described in s. 5.3.2.
- **No Pre-attribution Emergency Financing:** until attribution is demonstrated, affected parties (the “potential victims”) bear their own costs in response to any incidents.

5.1.4 Ransom Costs and Payments to Attackers Ineligible for Compensation

- Covered costs “directly resulting from an incident” under s. 5.1.1 do not include any payment(s) made to a cyber attacker or other “antagonistic party”.
 - For instance, a person that deceptively requests a ransom from the victim, despite knowingly not having been the entity that engaged in the criminal act, is an antagonistic party under this regulation.
- Examples of payments to attackers and antagonistic parties which are ineligible for compensation include:
 - (a) ‘Ransomware’ and direct ransom payment;
 - (b) Extortion payments;
 - (c) Any other money sent in exchange for:
 - (i) Restoring access to a victim’s own systems and database;
 - (ii) Preventing public disclosure of the incident;
 - (iii) Returning a victim’s own data;
 - (iv) Agreeing to cease or stop an ongoing cyber attack.
- The ineligibility of coverage under s. 5.1.4 is not to be read as impugning upon the judgement, prudence, or good faith of victims who make payments to attackers.
 - In some cases, paying an attacker may indeed be a victim’s most economically rational private decision.
 - This ineligibility is only to imply that where the Pool’s purpose is to mitigate immediate economic harms to victims, subsidizing criminal extortion is beyond its remit.

5.2 Return of Funds

5.2.1 Rolling 5-year Window

- Starting in year 5, a Pool Participant will be eligible to withdraw an amount (the “**Refundable Contribution**”):
 - (a) Equal to the *initial value* of her base contribution in the first year (year 0),
 - (b) Less the cost of any payouts made from incidents directly attributable to them,
 - (c) Plus any interest accrued on the *remaining amount* of her Base Contribution, after subtracting the cost of payouts, since that time.

■ The amount following from this process is given by the “**Capital Release formula**” below:

$$(\text{Base Contribution}_{\text{Year } t-5} - \text{Incident Payouts}_{\text{Year } t-5}) + \text{Interest}$$

5.2.2 Payouts Exceed Refundable Contribution (‘Incident Deficit’)

- The information herein s. 5.2.2, describes only the case regarding a **participant-attributable** loss to the Pool:
 - Where that loss exceeds the Participant’s own Base Contribution to the Pool for that year, and the Participant has not yet repaid the costs attributable.
- (a) Costs, obligations, and liabilities acquired by the Pool from incidents directly attributable to a Pool Participant are ‘participant-attributable’.
- (b) Where payouts attributable to a Participant’s exceed the implicated Participant’s refundable contribution for some year t , the difference shall be known as the Participant’s “**incident deficit**”:

$$\text{Payout} > \text{Refundable Contribution} \rightarrow \text{Incident Deficit}$$

- (c) Where a Participant has an active (non-zero) incident deficit:
 - (i) The incident deficit shall carry forward, and
 - (ii) The standing amount of the deficit will be deducted against all future Capital Releases otherwise payable to the Participant,
 - (iii) *Until* the Participant’s (attributable) incident deficit has been repaid.

5.2.2A Payouts Exceed Pool Balance (“Deficit”)

- In the more general case that payouts from an incident exceed the Pool’s entire Balance altogether—this is an ordinary “Deficit”.
- Ordinary deficits are addressed separately under s. 6.4.

5.2.3 Structuring of Incident Deficits (Solvency)

- Outstanding incident deficits are structured as a continuing **debt** owed by the implicated Participant(s) to the Pool.
 - Structuring deficits as debt creates a second layer of incentives by exposing shareholders and creditors to financial consequences that induce market discipline.
 - When a participant takes on too much risk and goes bankrupt, because shareholders are residual claimants, bondholders typically take priority, creating a reason for investors (shareholders) to prioritize transparency, alongside safety.
 - If two labs offer roughly the same returns, investors should prefer the one with lower risk. If the two labs present roughly the same risks, but only one of them communicates it, investors are likely to prefer the one whose risks are better quantifiable—given that understated risks could eventually manifest as debt that reduces the value of their investments.

5.2.4 Continuing Liability for Incident Deficits

- A Participant that leaves the Pool remains liable for repaying all outstanding debts incurred from incident deficits.

5.3 Attribution for Incidents

- An affected party (“potential victim”) who submits a claim to the Pool shall be known as a “Claimant”.
- Upon receiving notice of a claim, the Pool Administrator must inform all Participants of the claim, and provide all information from the Claimant available at the time.

5.3.1 Submitting a Claim

- Claimants are not required to prove causal attribution before submitting a claim.
- To submit a claim, Claimants must provide a credible report (“**Initial Report**”) that:
 - (a) demonstrates some evidence for believing that one or more Participants are implicated;
 - (b) describes their reasons for relying on the report;
 - (c) discloses, in full, their relationship to the Party authoring the report (the “**Author**”), including any dealings and business relationships, past or present, *where the report was commissioned by a third party*;
 - (d) provides all available information pertaining to the incident up to the time of submission.

5.3.2 Demonstrating Causal Attribution

- Causal attribution may be demonstrated using any reliable evidence, including evidence from:
 - (a) The affected entity or third-party cybersecurity contractors;
 - (b) Internal investigations conducted by Pool Participants;

- (c) Investigations by law enforcement and other, external regulatory bodies, and;
- (d) Independent technical investigators.
- The Pool Administrator’s role is not to independently decide whether a report submitted under s. 5.3.1 is true, or if the implicated Participant(s) caused the incident.
- The Pool Administrator need only determine if the evidence from a report submitted under s. 5.3.1 provides a reasonable and credible basis to suspect the implication of one or more Participants.

5.3.3 Approval of Claim

- Where the threshold under s. 5.3.2 has been met, the Administrator must approve the Claim.
- Claims approved are immediately eligible for payout, following submission of a final, attribution report (“**Final Report**”) containing:
 - (a) A signed statement by the Author of the Initial Report, where commissioned in-house or by third-party experts, attesting that their investigation was conducted in good faith and that the information in the report is true and complete, to the extent of their knowledge;
 - (b) A complete enumeration of all cybersecurity-related costs incurred directly as a result of the incident, covering the 30 day interval following from the date of first discovery, pursuant to s. 5.1.
 - (i) Costs borne by the approved claimant (“**victim**”) after this time are not eligible for compensation through the Pool.
 - (ii) Costs borne after the 30 day interval, including civil suits from the victim’s own shareholders, customers, or other entities for the incident, may be recovered through ordinary litigation (“**additional litigation**”) against the Participant(s).
 - (iii) When notified of additional litigation, the Administrator must provide all information previously collected from the incident under s. 5.3 if requested.
 - (c) A signed statement by the victim (approved Claimant) attesting to the veracity of the damages claimed in the Final Report under s. 5.3.3(b).
 - (d) A digital copy of all data gathered, relating to the incident, up to the submission of the Final Report.

5.3.4 Attribution Where a Model Is Used by a Third Party

- Incidents caused by a third-party’s use of a Participant’s model(s) do not, alone, prevent Causal Attribution to a Participant.
- The Pool is responsible for all Claims arising from incidents caused by an Implicated Participant’s model(s), regardless of whether the model was used by a third party.
- No part of this regulation precludes an Implicated Participant from later recovering costs, damages, or pursuing any other claims, resulting from the incident against the third party.
 - Any ongoing litigation between Parties arising from an incident described herein s. 5.3.4 shall not delay or reduce compensation otherwise payable to an approved Claimant.

5.3.5 Where Incident Is Addressed Before a Claim Is Made

- Should an incident be discovered and addressed by a Participant before a Claim is made, this does not affect a victim's right to later make a Claim for covered costs.
- An incident is considered "addressed" where a Participant has:
 - (a) Informed the Pool Administrator and affected parties of its discovery of the incident;
 - (b) Disclosed all relevant information to the Pool Administrator and affected parties necessary to mitigate the incident and document any resulting costs;
 - (c) Satisfied itself that the nature of the threat is no longer on-going and that the cause has been fully identified and will not recur again;
 - (d) Satisfied itself that it has identified all potential victims of an incident, including those beyond the immediate, affected Parties first informed under s. 5.3.5(a);
 - (e) Provided a signed affidavit affirming all of the above.
- An incident is considered "discovered" where a Participant has been made aware of an incident, but any one of the requirements for considering an incident addressed have not been met.
- Actions taken under s. 5.3.5, including prompt discovery, disclosure, and proactive remediation of incidents, may be considered in making a determination of a Participant's due diligence.

5.4 Causal Attribution Disputed by Participant

- An Accused Participant named in a Claim under s. 5.3.1 may submit evidence contesting the attribution ("**Statement of Dispute**").

5.4.1 Statement of Dispute

- A Statement of Dispute must contain:
 - (a) A complete disclosure of all information gathered internally by the Accused Participant and relied upon in disputing the attribution;
 - (b) An explanation of why the Participant believes that the information provided under s. 5.4.1(a) is complete and total;
 - (c) A signed statement agreeing to provide any further information later requested by the Claimant or their agent, with any redactions subject to the Approval of the Administrator.
 - (d) A signed affidavit agreeing to withhold any punishment, retaliation, or punitive actions against an employee of the Participant who submits information regarding the incident under s. 5.5.

5.4.2 Dispute Not to Delay Payout

- A dispute of attribution shall not otherwise stay or delay any payout from a Claim that has been approved by the Administrator.

5.4.3 Where Statement of Dispute Filed Before Final Report

- Where an accused Participant submits a Statement of Dispute before the Claimant's Final Report, the Claimant must:
 - Address the countervailing information and arguments provided by the accused Participant in their Final Report, or,
 - Provide evidence to the Administrator that doing so would create undue hardship or an unreasonable financial burden for the Claimant.
 - Where the Administrator accepts the claim of undue hardship, the Claimant is allowed to only partially address the evidence in the Statement of Dispute.
 - The Accused Participant may offer to cover the costs of any additional technical investigation borne by the Claimant necessary to fully address the Statement of Dispute.

5.5 Whistleblower Protection

- Pool Participants must provide employees with a confidential channel to submit any relevant information, relating to an incident or claim, *directly* to the Administrator.
- Pool Participants must agree to withhold retaliation against any employee(s) making good-faith disclosures under s. 5.5.

5.5.1 Evidence of Retaliation

- Evidence of retaliation may be considered in determining whether a Participant acted with due diligence.

5.6 Intentional Double-Counting

- An initial artifact under this scheme is the appearance that, in certain cases, an implicated Participant is penalized twice for the same incident.
- For example, suppose both the Pool Balance and Reserve Requirement are \$10 million. Company B then causes an incident requiring a \$10 million payout.

Step 1: The Pool compensates victims

- The Pool pays \$10 million to affected parties ("victims").

- Its Balance falls to \$0, creating a \$10 million Deficit.

Step 2: Participants replenish the Pool

- No agreement is reached within the 60 day window.
- The Statutory Backstop kicks in, and Participants collectively pay \$10 million through their respective Deficit Assessments.
- Now, the Pool's Balance consequently returns to its \$10 million Reserve Requirement.

Step 3: Company B's refundable capital is later reduced

- Under the Capital Release Formula, the \$10 million payout attributable to Company B is then deducted from future releases before any remainder can be paid out.
- The Pool may therefore accumulate funds **above** its \$10 million Reserve Requirement.

Part 6: Pool Balance and Minimum Reserves (“Reserve Requirement”)

6.1 Formula Terms and Notation

Let:

R_t = Reserve Requirement in year t

B_t = Reference Pool Balance in year t

D_t = Pool Deficit in year t

Where:

Equation 1: The “Reserve (Requirement) Formula”

$$R_t := R_{t-1} + \text{Baseline Contributions}_t + \text{Investment Returns}_t - \text{Baseline Capital Releases}_t$$

Equation 2: The “Balance Formula”

$$B_t := B_{t-1} + \text{Base Contributions}_t + \text{Investment Returns}_t \\ - \text{Capital Releases}_t - \text{Incident Payouts}_t + \text{Replenishment Payments}_t$$

6.1.1 “Base” vs. “Baseline” Distinction in Balance & Reserve Formulas

- For the purposes of the Reserve Requirement Formula, “**Baseline**” always refers to the amounts under the ‘**no-incident counterfactual**’.
- Because the Pool is designed such that realized risk allows the Pool Balance to push above its baseline reserve requirement:
 - Any money that enters or remains in the Pool, because of an incident, is excluded from the Reserve Requirement, and,
 - Is used to increase the Pool’s *Actual Balance* above its baseline amount instead (see s. 6.2.1).

6.1.2 Investment Returns Excluded from No-Incident Counterfactual

- The Reserve Requirement’s no-incident counterfactual applies only to **policy-determined components** in the formula (Equation 1), meaning:
 - Contributions, and
 - Capital Releases.
- For the purposes of Section 6, because investment returns depend on real-world market conditions:
 - Investment returns are to be treated as an accounting adjustment, reflecting the Pool’s actual portfolio performance, and
 - Investment returns are always calculated, under the formula, at their current, realized value.

6.1.3 “Incident Payouts” in the Balance Formula

- For the purposes of Equation 2, “Incident Payouts” always refers only to the portion of a payout funded from the Reference Pool Balance.
- Any amount financed through a designated liquidity facility is to be excluded and tracked separately as an Outstanding Liquidity Draw pursuant to s. 6.5.3.

6.1.4 “Replenishment Payments” in the Balance Formula

- For the purposes of Equation 2, “Replenishment Payments” always refers only to the portion of payments credited to the Reference Pool Balance.
- Any amount applied to repay an Outstanding Liquidity Draw is to be excluded from the Balance Formula and subtracted instead from the amount owing to Outstanding Liquidity Draws.

6.2 Pool Balance

6.2.1 Actual Balance

- The actual value of the assets in the Pool at any given instant is the Pool's '*Actual Balance*':

$$B_{\text{actual}} = \text{Value of Pool}_{\text{now}}$$

- The 'Actual Balance' is not the value of interest used in this document hereof.
- Whenever "Balance" appears in this text, it refers to the "**Reference Pool Balance**".

6.2.2 "Balance"

- The **Reference Pool Balance**, or "**Balance**", is the Pool Balance as calculated by Equation 2 under s. 6.1.
- This distinction arises because the usage of "Balance" in this paper *always* refers to the defined balance used by this policy's mechanisms, *not* the running total of the Pool at every moment.

6.2.3 Temporary Financing Excluded from Reference Pool Balance

- Temporary financing obtained through a designated liquidity facility pursuant to s. 7.4, is:
 - (a) To be tracked separately, and,
 - (b) *Always* excluded from the Reference Pool Balance (the "Balance").

6.2.4 Balance to Carry Over Cumulatively

- In any year t , the "Balance" will be the amount carried over from the previous year,
 - Plus the current year's mandatory base contribution from each Participant,
 - Plus investment returns,
 - Minus any Capital Releases (withdrawals) from eligible Participants,
 - less the cost of any payouts made, and,
 - Plus any Replenishments.

6.2.5 “Replenishment Payments”

- Replenishment Payments (**‘Replenishments’**) are any payments made by Pool Participants towards a Pool Deficit.
- Replenishment Payments may be applied to:
 - Restore the Balance towards the Reserve Requirement; or
 - Repay any Outstanding Liquidity Draws.
- Replenishment Payments encompass both:
 1. **Voluntarily Negotiated Replenishment**
 - e.g. Participants coming to a negotiated agreement a week after an Approved Claim,³² and paying back the Pool an amount as each of them has agreed.
 2. **Statutory Deficit Assessments**
 - e.g. Mandatory payments authorized by the Administrator under the statutory back-stop, forcing Pool Participants to pay down a deficit.

6.3 The Reserve Requirement

For the first year (Year 0), the Pool’s **“Reserve Requirement”** is initialized to be the Pool’s Balance at that time:

$$R_{t=0} := B_{t=0}$$

6.3.1 “Reserve Requirement”

- The Reserve Requirement is the minimum amount of money that the Pool must maintain across all subsequent years to ensure sufficient liquidity for payouts from future incidents.

6.3.2 ‘No Incident’ Counterfactual

- The Reserve Requirement follows from the ‘no incident’ counterfactual—the valuation of the Pool under the ideal condition where:
 - (a) no incidents occur, and
 - (b) the Pool grows at its natural rate under ordinary contributions, Capital Releases.
- With respect to calculating the no incident counterfactual, investment returns are excluded and treated separately as a realized empirical value pursuant to s. 6.1.2.

³²(well before the 60-day window)

6.3.3 Balance Not To Fall Below Reserve Requirement

- The Pool's Balance **cannot fall under the** Reserve Requirement.

$$B_t \geq R_t$$

6.4 Deficit

- The Pool is considered to be “**in deficit**” whenever either:

(a) its Balance falls below the Reserve Requirement;

$$B_t < R_t$$

(b) Or, it has made Outstanding Liquidity Draws that have not yet been repaid.

$$\sum_{i=0}^t \text{Liquidity Draws}_i - \sum_{i=0}^t \text{Liquidity Repayments}_i > 0$$

- The Pool's Deficit is defined as:

$$\text{Pool Deficit}_t = \max(0, R_t - B_t) + \text{Outstanding Liquidity Draws}_t$$

6.5 Outstanding Liquidity Draws

- The Reserve Requirement, and the Pool replenishment mechanism (Deficit Assessments) do not alone provide a means for covering the cost of arbitrary payouts from the Pool.

6.5.1 Financing from Designated Liquidity Facility

- Given that the Pool is intended to help preemptively prepare for costs from expensive, catastrophic ‘tail risk events’,
 - Wherever payouts exceed the value of the Pool's prefunded assets,
 - Temporary financing from a designated liquidity facility *must* bridge the gap until Participants replenish the Pool.

6.5.2 “Liquidity Draw”

- The liquidity draw is the amount borrowed from a designated liquidity facility to fund payouts in the event of costly tail-risk events that exceed the Pool's available assets.

$$\text{Incident Payouts}_t > \text{Pool Balance}_t$$

- Where a payout exceeds the Pool's available liquid assets, but not the true Balance, the Administrator is nonetheless authorized to draw upon the designated liquidity facility.

6.5.3 “Outstanding Liquidity Draws”

- Where the Pool has not yet repaid the temporary financing provided by a liquidity facility under s. 6.5.1, this amount, still owing, is the Pool’s “**Outstanding Liquidity Draw**”:

$$\text{Outstanding Liquidity Draws}_t := \sum_{i=0}^t \text{Liquidity Draws}_i - \sum_{i=0}^t \text{Liquidity Repayments}_i$$

- Ordinarily, where the Pool’s Balance is large enough to cover all payouts made during the period, this term (“Outstanding Liquidity Draws”) is zero.
- The Outstanding Liquidity Draws term matters whenever an unpaid (outstanding) obligation exists, not merely at the instant where a payout exceeds the Balance. This means the term may continue to matter even after an immediate cash shortage is over.
- For instance, if, following an incident:
 - The Pool restores its Balance up to the Reserve Requirement,
 - but still owes \$5 million to the liquidity provider,
 - The Pool will *still be considered* to sit at \$5 million **in Deficit**.

Example

Let the Pool contain \$15 million in year t . An incident involving a \$20 million payout occurs.

- The Pool borrows \$5M, and pays out the full \$20M. The Pool now has:
 - Balance = \$0
 - Reserve Requirement = \$15M
 - Debt to Liquidity Facility = \$5M
- Thus the Pool needs \$15M to restore its reserves:

$$R_t - B_t = \$15M$$

- However to fully return to normal, the Pool actually requires \$20M:
 - \$15M to restore its Balance to the Reserve Requirement, and
 - \$5M to repay the liquidity facility.
- Thus the \$5 million remains in the Pool Deficit until the facility is repaid.

6.6 Repayment of Outstanding Pool Deficit (60 Day Window)

6.6.1 Negotiated Private Replenishment

- At any time when there is an outstanding deficit—for instance, following a payout from an incident—a **sixty day window** begins (from the first day the Pool is in deficit).
 - All Participants in the Pool are jointly responsible for paying down the deficit within the 60 day period.
 - Participants are free to negotiate their individual Replenishment obligations amongst themselves.
 - For instance, if Company A was responsible for 100% of a given year's deficit, but can cover only 50% of it without becoming insolvent, the other participants in the pool may agree to cover the remainder, contingent on some compensation or repayment schedule agreed by all parties.
- If the parties (Participants) are unable to reach a private agreement within 60 days, repayment of the deficit will occur immediately, with individual obligations structured as defined under the statutory backstop.

6.6.2 Statutory Backstop (Deficit Assessment)

- If the Pool remains in deficit at the end of the 60 day window, the Pool's administrator shall, without delay, issue a mandatory payment (the "**Deficit Assessment**") from all participants.
- Each Participant's Deficit Assessment is to be automatically determined *pro rata* by revenue, meaning each participant is responsible for paying an amount equal to:
 - The same percentage of the Pool Deficit as its annual revenue represents proportionally among the combined annual revenue of all Pool Participants.

$$\text{Deficit Assessment}_i = \text{Pool Deficit} \times \frac{\text{Participant Revenue}_i}{\sum_{j \in \text{Participants}} \text{Participant Revenue}_j}$$

Example

Let the current Pool Balance and Reserve Requirement in year t be \$15 million.

- Following a cybersecurity incident caused by Participant A's models, the payout to affected parties is \$10 million.
- The Pool's Balance is now \$5 million. The Reserve Requirement, unchanged, remains at \$15 million for year t .
- Subtracting the Balance from the Reserve Requirement produces a deficit of \$10 million.

Participant	Revenue	Share of Total Pool	
		Revenue	Deficit Assessment
A	\$100M	23.81%	\$2.38M
B	\$120M	28.57%	\$2.86M
C	\$200M	47.62%	\$4.76M
Total	\$420M	100%	\$10.00M

6.6.3 Without Prejudice

- No statement, communication, representation, disclosure, or concessions made during dealings under s. 6.6.1 should be taken to constitute an admission of guilt, liability, or otherwise prejudice a party's rights or position in subsequent litigation.

Part 7: Sovereign Liquidity Backstop

7.1 Purpose of the Sovereign Liquidity Backstop

- The purpose of the Sovereign Liquidity Backstop is to:
 1. ensure that States supply the emergency liquidity needed to fulfill the spirit of the Pool's Purpose under Section 3, even as industry remains the ultimate loss-bearer;
 2. Improve price discovery and economic coordination, by ensuring that financial risks created by frontier labs (Pool Participants) are reflected accurately to investors in their business costs, rather than borne by third parties as negative externalities to the wider economy.

7.2 Nature and Definition of the 'Scheme'

- Where one or more States implementing this policy coordinate on implementation ("Participating Jurisdiction"), the joint implementation of the policy shall be known as the "Scheme".

7.2.1 Sovereign Responsible for Harms Outside Jurisdiction

- The Sovereign, under whose jurisdiction the risk-producing activities from Pool Participants occurs, must:
 - supply temporary liquidity for harms caused resulting from such activity,
 - including harms that occur outside its jurisdiction.
- This is an acknowledgement of the permeable nature of geographic borders on the modern internet, as well as the cross-border nature of cybersecurity threats.

7.2.2 Conditions of Ratification

- States participating in the Scheme must specify explicitly, either in:
 - (a) The implementing statute, or
 - (b) Delegated legislation, if implementation is pending Royal Assent or Presidential Approval,

that domestic Pools must compensate qualifying victims without discrimination of nationality, location, or geography.

7.3 Sovereign To Authorize Lending in Event of Shortfalls

- The Sovereign of a Participating Jurisdiction must enact legislation establishing the statutory authority required to meet the liquidity obligations defined under s. 7.4.

7.4 Domestic Designated Liquidity Facility

- Every jurisdiction participating in the Scheme must guarantee that its domestic Pool is capable of paying approved claims in full, notwithstanding the insufficiency of its domestic Pool's assets.

7.4.1 Structure of Domestic Liquidity Backstop

- The Sovereign of a participating jurisdiction may decide the structure and institutional form of their domestic liquidity backstop, such as whether:
 - (a) The State's Treasury, or equivalent authority, provides the standing credit facility directly;
 - (b) Private lenders provide the domestic Pool with a guaranteed standing credit line, backed by the State;
 - (c) Any other arrangement which is sufficient for meeting its obligations under the Scheme.
- This is intended to provide Participating Jurisdictions with the flexibility to implement the Scheme in a manner compatible with their own domestic legal, political, and institutional traditions.
 - For instance, States with a strong commitment to free markets and minimal economic intervention may prefer implementation of the Scheme through private financial intermediation.
 - Conversely, States with strong social democratic traditions may elect to use direct public provision.

7.4.2 Sovereign Fallback

- However a Participating State's Domestic Liquidity Backstop is structured, wherever the designated liquidity facility chosen under s. 7.4.1 is unable to advance the required funds, when due, the Sovereign must advance the outstanding shortfall directly.

7.5 International Coordination of Liquidity Backstops

This Scheme does not establish a 'common international compensation fund' nor require that Participating Jurisdictions guarantee liquidity to another's domestic Pool. Instead, States participating in the Scheme agree to common minimum standards and retain responsibility for the risks generated by Participants in their domestic Pool.

7.5.1 Common Minimum Standards

- Participating Jurisdictions must adopt common minimum standards to fulfill the spirit of the policy, as defined in the Purpose of the Pool under Section 3.
 - This includes standards governing the operation of the Pool and the sufficiency of Domestic Liquidity Backstops.

7.5.2 Responsibility To Follow from Source of Externality

- Responsibility for the payout of Approved Claims are to follow the Implicated Participant(s), rather than the jurisdiction where the harm occurred.
- Claimants located outside the Participating Jurisdiction may submit a Claim against the Accused Participant's domestic Pool with the same rights and privileges as a domestic Claimant.

7.5.3 Participating Jurisdictions Individually Responsible

- Each Participating Jurisdiction is responsible for ensuring its domestic Pool meets the liquidity requirements under the Scheme.
- No State shall be responsible for financing a shortfall arising from the activities of a Pool Participant regulated by another Participating Jurisdiction.

7.5.4 Cooperation and Information Sharing

- Participating Jurisdictions are encouraged to further build on the Scheme by cooperating and exchanging information to fulfill the spirit of the policy, including:
 - (a) Implementing additional domestic legislation, if necessary, to facilitate the timely processing of cross-border claims;
 - (b) Sharing best practices on administering regulations, and;
 - (c) Engaging with each other in open dialogue or backchannel communications to settle disagreements in good faith, where they believe the Scheme is inadequate or inequitable.

Appendix

Appendix A. Rationale

A.1 Why a rolling 5 year window instead of a solvency criteria based on risk assessments of plausible financial costs?

This design is intended to optimize partly for robustness to influence, over actuarial efficiency. Imagine if the policy stated:

“The Pool’s administrator shall determine capital contribution requirements pursuant to reasonably foreseeable, future AI cybersecurity losses.”

- Now, Anthropic argues that new interpretability tools have reduced the expected severity and rate of future incidents by 64%, so a smaller Pool is sufficient, and annual contributions should fall.
- An external AI safety organization submits that contrary to Anthropic’s report, they assessed that autonomous capabilities have increased 3-fold, and the Pool’s reserves must increase 300% to avoid insolvency under likely payouts.
- A consulting firm hired by frontier labs submits a 100-page report contesting some arguments, but backing others.
- A separate report produced jointly by top research universities completely disputes the assumptions made in the 100-page report.
- The government changes following an election and it appoints a new administrator who chooses a new methodology that produces new figures that now favours one side or the other.
- ...And now we have the ingredients for a court challenge.³³

The allegedly technocratic variable, R = Required Pool Reserves, has now opened the entire design up to subjective assessments of risk and thereby ‘gaming’ from participants by making it a complicated function of:

$$R = f(\text{models, regulatory discretion, assumptions, political pressure, ...})$$

This proposal mitigates the risk of parties ‘gaming the system’ by selecting a mechanism insulated from these subjective questions. By hard-coding the “**Contribution Formula**” as:

$$\text{Contribution}_{\text{year}} = \max(0.01 * \text{Revenue}_t, \$5\text{m})$$

And by hard-coding the lock-up period to be 5 years, with no discretion, all parties know the rule before making decisions.

An accelerationist does not get to make a case for reducing contributions because they predict safer models. The cost is that safety advocates do not get to ask for an increase either because

³³And one best hopes there are no appeals.

they predict catastrophe. Either of them may ultimately turn out to be right or wrong; neither prediction gets to alter the *ex ante* mechanism.

Importantly, such a mechanism is not merely ‘fair’ to businesses, it is also predictable. Thus, a secondary benefit of a non-discretionary mechanism is that it avoids, in as much as possible, the regulatory trap of a policy that creates greater uncertainty for businesses and investors.

A.1.1 The Null Hypothesis Argument (Why Fixed Base Contributions?)

A second question might be why the Contribution Formula uses hard-coded (fixed) amounts that seemingly treat AI-related cybersecurity risks as a constant. This proposal does not assume that such risks are constant.

Instead, the Contribution Formula is intended as a ‘default rule’, constructed deliberately to account for the specific circumstances that motivate the use of a strict liability regime: wherever the risks from some given activities are likely to be significant, but where the true magnitude and distribution of future losses remain unknown.

The policy logic underlying this choice is hence analogous to a null hypothesis in science. Absent (sufficiently) strong reasons to alter the regulatory scheme from its baseline, the existing obligations of Participants under the Contribution formula remain in place.

A.2 Why treat incident deficits as debt?

The Capital Release Formula is intended to ensure that Participants bear some individual responsibility for the costs attributable to them from incidents by deducting these costs from a Participant’s refundable amount (*i.e.* their Capital Releases).

The problem arises when the cost of an incident exceeds an Implicated Participant’s amount of refundable capital available. In such a case, merely deducting the losses borne by the Pool from the Participant’s (future) Capital Releases may still leave the remainder of the loss unaccounted for.

In effect, the Participant’s liability would be capped at however much capital it happened to have deposited into the Pool until this time.

By treating any outstanding, Participant-attributable amounts as ‘incident deficits’, this proposal avoids this possibility. Since incident deficits are debt, the unpaid amounts remain attached to the Implicated Participant. Future Capital Releases may then be applied to reduce the outstanding debt, while the Implicated Participant remains free to repay their debt directly.

A.3 (Perverse Incentive) If victims know that someone else is paying the costs, what stops them from choosing the \$8M response when a \$2M response would have been perfectly adequate?

One concern is that the policy’s objective of ensuring that victims are compensated fully could create a blank cheque for unlimited cybersecurity defense spending in the wake of incidents. However, the current mechanism, as proposed, provides a good check against this.

A.3.1 Reimbursement is uncertain *ex ante*

In the immediate aftermath of an incident, victims do not know whether the attack will ultimately be attributed to a Pool Participant. As such, victims are incentivized to spend only as much as is believed to be required, to secure their infrastructure following an attack.

At the same time, market incentives should push victims not to under-spend, as doing so may expose them to legal liability from their own shareholders and customers.

A.3.2 Opportunistic overspending $\neq$ overspending caused by uncertainty

If a company affected by a cybersecurity attack spends \$8M because no one in the company has the expertise to determine if the attacker still has persistence, or what systems remain compromised, those costs—resulting directly from the frontier labs’ lack of transparency with the public—become part of the incentive structure for labs to communicate better about their practices and internal developments.

For instance, if a lab (“Participant”) is developing a powerful new model internally that may escape containment and cause harms, even before it is released:

- **The Participant** is incentivized to make a public announcement about what it is doing, so that members of the public know where to direct reports about potential incidents in a timely manner, to reduce related costs to both the victim(s) and the Pool.
 - Courts may also later use information about these actions, or the lack thereof, in determining due diligence under application of the Strict Liability standard.
- **Other Pool Participants** are incentivized to inform regulators about potential developments, like new models, that other Participants have not disclosed, given that they will bear some of the costs

A.4 Why the Attribution Mechanism Does Not Require Establishing ‘Proof’

A regime may nominally be described as a “strict liability regime”. But functionally, if Claimants must first be able to finance a prolonged legal dispute over causal attribution before being compensated for harms, then from a victim’s perspective, this design continues to preserve a major, long-standing, practical defect of fault-based liability when used as a restorative justice mechanism.

A.5 (Perverse Incentive) Strategic Claiming: If parties know payouts will proceed even if attribution is eventually overturned, what stops frivolous Claims?

One concern is that the combination of: **(1) the low causal attribution threshold³⁴** and **(2) payouts proceeding even when under dispute** create a perverse incentive for parties to file false claims, since they know that payouts will proceed even if attribution is eventually overturned.

This is a genuine problem which may provoke the question: why is there no explicit clawback provision in the regulation for incorrect attributions or harsh punishments for frivolous (false) claims?

³⁴Requiring only a reasonable basis for suspicion and not proof.

A.5.1 Dealing with Frivolous Claims

One reason relates to power dynamics. Under this regulation, the deterrence against frivolous claims is intended to be the inherent, structural power asymmetries between frontier labs and ordinary companies/persons.

Imagine you're an average company, and now you're making a decision about whether or not to lie about being hacked to try and gain some moderate amount of money from an AI frontier lab (whose revenues sit several orders of magnitude above your own).

Now, if criminality is anywhere near parity with intelligence, you would likely come to the realization quickly that fraud is typically most successful when conducted against a party that neither has the funds to litigate your lies, or the reputational interest to fight against every baseless claim, however minute.

Insurance fraud works largely because insurance companies process thousands of claims a day and cannot possibly scrutinize every one. Employees can submit fraudulent timesheets for over-time pay because middle managers do not have the ability to verify every item on a submission.

But here, where the targets of the fraud would be an AI frontier lab—*prima facie*, almost certainly some of the richest, best-resourced, highly-lawyered entities on the planet—the cost-benefit analysis would make filing such a fraudulent cybersecurity claim (e.g. against a company like Google) akin to taking a gamble on shoplifting from a store that sells security cameras.

For any potential, aspiring fraudster the expected value calculation amounts to:

- “If I succeed, I may get a moderate-to-high payout”,
- “If I fail, I face civil liability for fraud, criminal liability for perjury (over the signed attestation required under s. 5.3.3), reputational harm”,
- “Also I will probably almost definitely be caught because my victim is a company with a legal budget several multiples of my monthly revenue.”

In short, no rational actor takes that bet. The power asymmetry between the labs and potential Claimants would, in almost all circumstances, naturally generate its own structural deterrence mechanism.

A.6 Why Are Participants Responsible for Claims Caused by a Third Party's (Mis)use of Their Models?

Under this regulation, an Incident may be causally attributed to a Participant, so long as their models are involved, even if the proximate cause of the Incident was due to a third party's use of their models.

There are three reasons why this is a deliberate design of the proposal.

A.6.1 Simplicity and Relief for Victims

A strict liability regime is supposed to produce a regulatory environment that is simple to navigate and quick to respond. Because one objective of this policy is restoration/redress for victims, compensation should not be delayed simply because better-resourced, well-funded commercial parties dispute responsibility over the proximate cause of an Incident.

Additionally, allowing such an exception would create an inconsistency under the regulation, where:

- (a) disputes over attribution do not delay payouts from Claims,
- (b) but a payout may be delayed due to disputes over liability, even when proximate attribution (the third party) and ultimate attribution (the Participant) are not disputed.

A.6.2 Participant Uniquely Positioned To Address Failures

This proposal attaches liability to a Participant, even where an Incident arises from a third-party's actions because a second objective of this policy is deterrence. If frontier AI labs release models which uplift the capabilities of bad actors to cause harm, but they are held responsible only where they directly engendered said harm themselves, this would create incentives for labs to invest better into safety during training and testing, but not deployment. In effect, this would only solve half the problem.

Where the impacts to the wider economy are concerned, the overarching goal is not to punish Participants for what they have done, but to create incentives to change their behaviours so that harms do not occur—and where they do occur, historical evidence of good faith regulatory participation minimizes reputational harm to the AI industry as a whole.

Where AI development is concerned, frontier labs are the parties uniquely positioned to best identify and correct problems with their products, as well as generalize from lessons learned from failure modes revealed by incidents.

Every incident therefore gives a Participant information that can be used to prevent similar harms in the future, but only an incentive mechanism that holds them liable, even if caused by third parties, gives them a reason to act on every incident.

A.6.3 Why Third-Party Use Does Not Shift Ultimate Responsibility

In the case that a third-party's normal, everyday use of a Participant's model(s) causes a cybersecurity attack, the third party did not 'magically' create that failure mode. If an otherwise ordinary deployment reveals that a model will autonomously pursue unlawful actions, if not explicitly instructed against doing so, such an incident should be considered as revealing a pre-existing misalignment problem in the model.

From this, a secondary reason why third-party-caused incidents should not shift a Participant's responsibility is because this would effectively let frontier labs outsource both the costs of their mistakes, and the work of discovering their mistakes.

Put another way, while car manufacturers should use empirical evidence to determine the need for seatbelts, they should not rely on the public first getting into accidents to obtain empirical evidence for them.

A.7 (Double-Counting) Why Should Realized Risk from Incidents Push the Pool Above its Reserve Requirement?

The proposal contains two mechanisms which, read together, may superficially provide the appearance of double-counting when penalizing Implicated Participants for incidents: **(a) Replenishment Payments**, and **(b) Withholding of Capital Releases**.

However, the payments associated with each of these two mechanisms are intended to serve different roles:

- (a) Collective Replenishment following a payout is meant to restore the Pool to its minimum Reserve Requirement to ensure it remains capable of compensating victims of future incidents without delay;
- (b) Conversely, the withholding of an Implicated Participant's Capital Releases ensures that realized costs, attributable to a Participant, are reflected in the Pool's Balance as additional funds.

If amounts withheld from an Implicated Participant's Capital Releases are allowed to count towards restoring the Pool's Reserve Requirement, a costly, tail-risk incident could deplete the Pool, only for it to return to the same minimum amount required before the incident occurred.

As such, evidence of realized risk (from incident payouts) would produce no lasting increase in the Pool's Balance, weakening the proposal's role as a deterrence mechanism, even while still preserving its restorative function.

A.8 Explanation of the Pool Balance and Reserve Requirement Formulas

Recall the functional definitions of the Pool's *Balance* and *Reserve Requirement Formulas*.

$$R_t := R_{t-1} + \text{Baseline Contributions}_t + \text{Investment Returns}_t - \text{Baseline Capital Releases}_t$$

$$B_t := B_{t-1} + \text{Base Contributions}_t + \text{Investment Returns}_t \\ - \text{Capital Releases}_t - \text{Incident Payouts}_t + \text{Replenishment Payments}_t$$

One question is why, in calculating the Reserve Requirement_t (or "*R_t*") for any year *t*, "**Baseline Contributions**" & "**Baseline Capital Releases**" are used instead.

- This is to ensure that the arithmetic reflects the possible outcomes of the Pool under interactions with other mechanisms in this policy's design.
- In particular, the Reserve and Balance Formulas are constructed as such because they need to account for changes to the Pool's value from this policy's "**incident machinery**".
- In plain English, we mean the two mechanisms under this proposal which can cause the Pool's Balance to change as a result of incidents—hence, *incident machinery*:
 1. **The Update Formula**, which can make a (Implicated) Participant contribute more following an incident; and

2. **The Capital Release Formula**, through the Pool Administrator withholding money, due to applying the penalty of subtracting the cost of incidents paid out from a Participant's eligible Capital Release.

- **Notice:** these (the above Formulas) are both about *incident-caused changes*.
- But this is a problem, because the Reserve Requirement, R_t , is intended to represent the counter-factual where no incidents occurred.

Demonstration

For instance, after an incident, we want:

actual extra contribution $\rightarrow B_t$ increases

But what we do not want is:

actual extra contribution $\rightarrow R_t$ also increases

- The reason for this is simple; if any extra capitalization simply raises the 'floor' (*the Reserve Requirement*) together with it, this design loses the '**Double Counting**' mechanism ("**DCM**") described in Section 5.6.
 - Whose intention is to make it so that: *realized risk can always push the actual Pool above its baseline reserve requirement.*
- This, for instance, is the reason for the distinction between Capital Releases _{t} and Baseline Capital Releases _{t} in the Balance and Reserve Requirement Formulas respectively. It is because the intentional DCM depends on exactly this gap.

Example

Suppose after 5 years of membership in the Pool, **Participant A** in year t would have ordinarily been eligible to receive back a \$10 million ("\$10M") Capital Release.

This is the money they put down in the first year of the Pool.

1. However, in the first year, $t - 5$, the Participant was also implicated in an incident costing the Pool \$10.1 million.
2. Now, recall the Capital Release Formula:

$$\text{Capital Release}_{t+5} = (\text{Base Contribution}_t - \text{Incident Payouts}_t) + \sum_{t=0}^5 \text{Interest}_t$$

3. Since in year $t - 5$:
 - Participant A's base contribution was \$10 million.
 - But the Payout from incidents they were responsible for was \$10.1 million.
 - And \$10 million – \$10.1 million = **not a positive number**
4. If the Reserve Requirement Formula then *subtracts* that actual release,
 - Which, remember, was –\$0,

5. Then the Reserve Requirement, R_t , also stays \$10 million higher too.

But the whole system was designed so that money withheld due to incidents should make the *actual* Pool's Balance rise about the no-incident Reserve Requirement!

6. Thus, to get that actual result, mathematically, the Reserve Requirement Formula must behave as though ordinary Capital Releases happened. This means:

- So this means:

- Although in year 5, Participant A's withdrawal (Capital Release) of its year 0 funds were... well \$0 (there was no money for them to withdraw).

- The Reserve Requirement nonetheless:

- Subtracts the \$10M 'ghost withdrawal' that Participant A *could* have made, had they been on their best, exemplary behaviour.

- And hence:

Reserve Requirement_t → Reduced by \$10 million

- While the actual Balance becomes:

Balance_t → \$0

- That difference is what creates: $B_t \geq R_t$

In other words, this is the bookkeeping that mathematically keeps the Pool Balance above the Reserve Requirement's no-incident counterfactual.

Bibliography

- Anthropic. "Expanding the Cyber Verification Program." October 6, 2026. <https://www.anthropic.com/news/cyber-verification-program>.
- Anthropic. "Investigating Three Real-World Incidents in Our Cybersecurity Evaluations." July 30, 2026. Accessed October 7, 2026. <https://www.anthropic.com/news/investigating-incidents-cybersecurity-evals>.
- Anthropic. "Making AI Systems You Can Rely On." Accessed October 7, 2026. <https://www.anthropic.com/company>.
- Coase, Ronald H. "The Problem of Social Cost." *Journal of Law and Economics* 3 (October 1960): 1-44. <https://doi.org/10.1086/466560>.
- Epoch AI. "Epoch Capabilities Index." Published October 2025. Last updated October 6, 2026. Accessed October 7, 2026. <https://epoch.ai/eci>.
- Epstein, Richard A. "A Theory of Strict Liability." *Journal of Legal Studies* 2, no. 1 (1973): 151-204. <https://doi.org/10.1086/467495>.
- Frenkel, Sheera, Dustin Volz, and Dylan Freedman. "OpenAI Ignored Employees Who Warned It Wasn't Doing Enough About Security." *New York Times*, September 29, 2026. <https://www.nytimes.com/2026/09/29/technology/openai-warnings-security.html>.
- Gilles, Stephen G. "Negligence, Strict Liability, and the Cheapest Cost-Avoider." *Virginia Law Review* 78, no. 6 (1992): 1291-1375. <https://doi.org/10.2307/1073455>.
- Goldberg, John C. P., and Benjamin C. Zipursky. "The Strict Liability in Fault and the Fault in Strict Liability." *Fordham Law Review* 85, no. 2 (2016): 743-788. <https://ir.lawnet.fordham.edu/flr/vol85/iss2/16/>.
- Greenblatt, Ryan, Ajeya Cotra, and Hjalmar Wijk. "Brief Independent Investigation of Agents' Behavior, Reasoning, and Collaboration in the OpenAI / Hugging Face Hacking Incident." *METR*, August 26, 2026. <https://metr.org/blog/2026-08-26-openai-hugging-face-incident-investigation/>.
- Gwern. "Why Tool AIs Want to Be Agent AIs." *Gwern.net*. Published September 7, 2016. Last modified August 28, 2018. <https://gwern.net/tool-ai>.
- Jaeger, Walter H. E. "The Warranty of Habitability - Part II: Strict Liability versus Absolute Liability." *Chicago-Kent Law Review* 47, no. 1 (1970): 1-82. <https://scholarship.kentlaw.iit.edu/cklawreview/vol47/iss1/2/>.
- OpenAI. "Daybreak." Accessed October 7, 2026. <https://openai.com/daybreak/>.
- OpenAI. "How We Will Do Better for Australia." September 28, 2026. Updated October 4, 2026. <https://openai.com/index/how-we-will-do-better-for-australia/>.
- OpenAI. "OpenAI Charter." April 9, 2018. <https://openai.com/charter/>.
- Pichai, Sundar. "AI at Google: Our Principles." *Google*, June 7, 2018. <https://blog.google/innovation-and-ai/products/ai-principles/>.
- R v Sault Ste Marie (City of)*, [1978] 2 SCR 1299.
- Reuters. "MGM Resorts Says Regulators Probing September Cyberattack." February 23, 2024. <https://www.reuters.com/technology/cybersecurity/mgm-resorts-says-state-federal-regulators-probing-september-cyberattack-2024-02-23/>.
- Robinson, David. "I Quit OpenAI Because Its Culture Is Broken." *The Atlantic*, October 3, 2026. <https://www.theatlantic.com/technology/2026/10/openai-safety-team-resignation/688881/>.